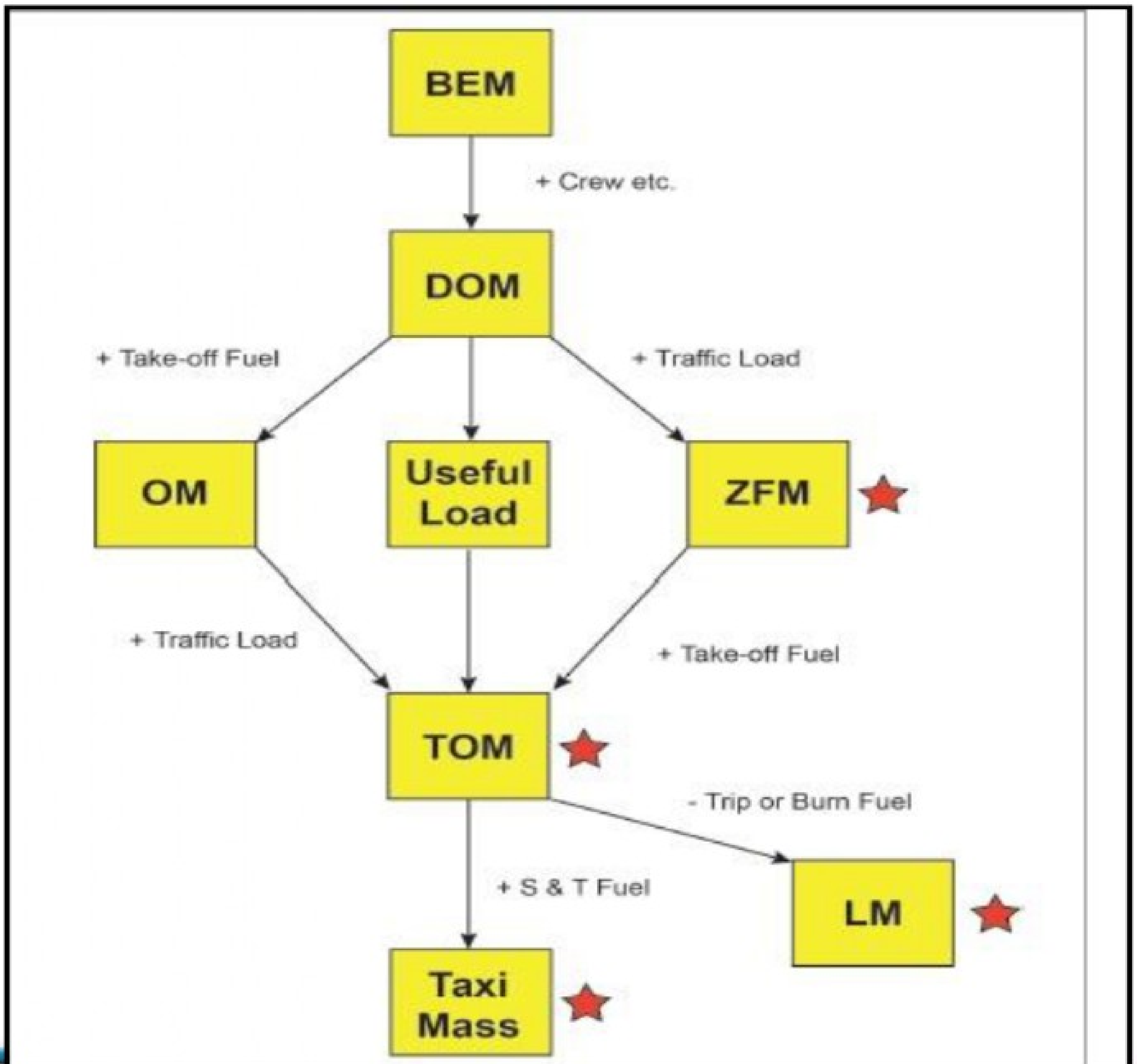

BASIC EMPTY MASS: - Is an aircraft's mass plus standard items such as:

- ✓ Unusable fuel and other unusable fluids
- ✓ Lubricating oil in the engine and auxiliary units
- ✓ Fire extinguishers
- ✓ Pyrotechnics
- ✓ Emergency oxygen equipment
- ✓ Supplementary electronic equipment
- ✓ The mass of the aircraft with all its basic equipment plus a declared quantity of unusable fuel and oil.

DRY OPERATING MASS (DOM)

The total mass of the aero plane ready for a specific type of operation **excluding all fuel and traffic load**. This mass includes items such as:

BEM + Variable Load Dry Operating Mass

1. Crew and crew baggage.
2. Catering and removable passenger service equipment
3. Potable water and lavatory chemicals.

TRAFFIC LOAD

- ✓ The total mass of passengers, baggage and freight including any non-revenue load

USEFUL LOAD

- ✓ The total of Traffic Load plus Useable fuel. (PAYLOAD + FOB)

FUEL ON BOARD

- ✓ Total Fuel = Trip fuel (Flt. Fuel) + Reserve

ZERO FUEL MASS (ZFM)

- ✓ Total mass of the Dry Operating Mass plus the Traffic Load.

MAXIMUM ZERO FUEL MASS (MZFM)

- ✓ The maximum permissible mass of an aero plane with no useable fuel. The mass of fuel contained in particular tanks must be included in the ZFM when specified in the aircraft flight manual limitations.

Take-Off Mass (TOM)

- ✓ The mass of an aero plane including everything and everyone in it at the start of the take-off run.
- ✓ DOM + TL + FUEL at take of run

Maximum Structural Take off Mass (MTOM) (Regulated TOW)

- ✓ The maximum permissible total aero plane mass at the beginning of the take-off run.

Performance Limited takeoff mass

- ✓ It is take off mass limited due to performance factors such as wet runway, high temperature.

Regulated take off mass

- ✓ It is lowest of take-off mass and performance limited take off mass

Maximum Structural Landing Mass (MLM)

- ✓ The maximum permissible total aero plane mass upon landing under normal circumstances.

Maximum Ramp Mass (Max Structural Taxi Mass)

- ✓ The maximum approved mass for commencement of ground maneuvers. A mass greater than the Maximum take Off Mass, to allow for fuel used in startup and taxi.
- ✓ Ramp mass = TOM + Fuel for startup + taxi

TO Calculate the Payload

RTOW

MLW +Flt. Fuel **Minimum Value**

MZFW+ FOB

Lowest of the three = A.P.S+F.O.B+PAYLOAD

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FUEL MASS POLICY

Trip Fuel

✓ The required fuel quantity from brake release at the departure airport to the landing touchdown at the destination airport is referred to as trip fuel. This quantity takes into account the necessary fuel for:

☞ Take-off
☞ Climb to cruise level
☞ Flight from the end of the climb to the beginning of the descent, including any
☞ step climb/descent
☞ Flight from the beginning of the descent to the beginning of the approach,
☞ Approach
☞ Landing at the destination airport

Contingency Fuel

✓ For enroute weather, lower flight level, deviation from planned route and flight level maximum would be 5% of trip fuel.

Alternate Fuel

✓ From destination missed approach to alternate.

Final Reserve Fuel

✓ The final reserve fuel is the minimum fuel required to fly for 30 minutes (Jet engine) at 1,500 feet above the alternate airport or destination airport, if an alternate is not required, at holding speed in ISA conditions
✓ For piston engine it is 45 minutes
✓

Additional Fuel

✓ Over and above trip + contingency + alternate + reserve

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- ✓ To fly enroute alternate when no destination available (Island hold fuel)

Extra Fuel

- ✓ Extra fuel is at the captain's discretion

Isolated Airport Procedure

- ✓ For such an airport, there is no destination alternate. The regulatory takeoff fuel quantity must include:
- ☞ Taxi fuel
- ☞ Trip fuel
- ☞ Contingency fuel, as calculated in the standard fuel policy
- ☞ Additional fuel: This quantity must be higher than the quantity necessary for a two-hour flight at cruise rating above the destination airport; final
- ☞ reserve fuel is included in this amount
- ☞ Extra fuel, is at the captain's discretion

Block fuel = All

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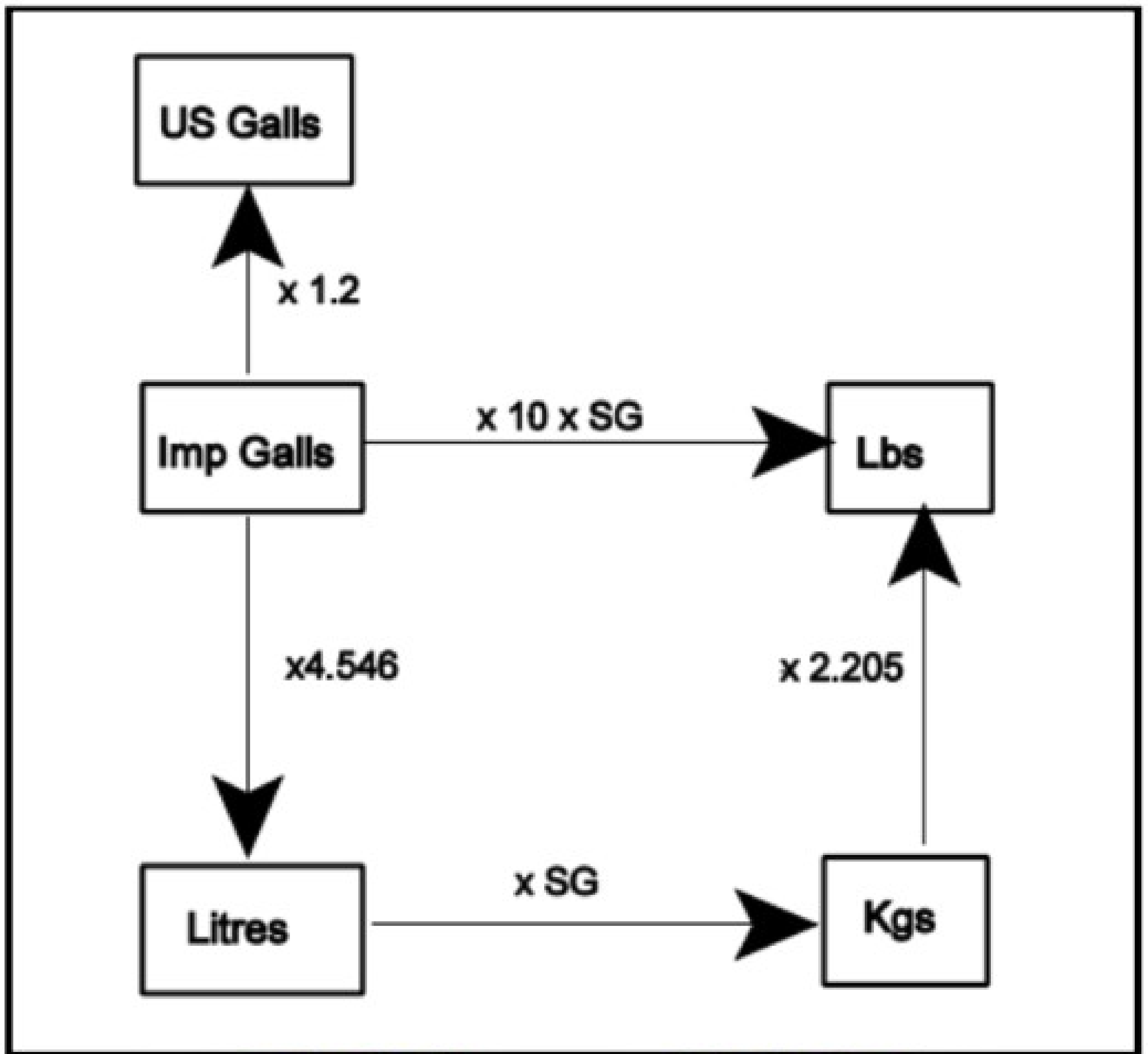
VOLUMETRIC AND MASS CONVERSIONS



- ✓ Specific Fuel Consumption (SFC): is defined as the weight of fuel (Kgs) required to fly one ground nautical mile (gnm). The lower the SFC the more economical the flight.

To calculate Kgs per gnm:

$$\mathbf{SFC} = \frac{\textbf{Fuel Flow (Kgs/hr)}}{\textbf{Groundspeed (gnm/hr)}}$$

- ✓ Thus an aircraft flying at a groundspeed of 400 kts using 6000 Kgs/hr would have an SFC of 6000/400=15 kgs per gnm. (Although Kgs are normally used, other units can be used e.g. liters, gallons, lbs etc.).

- ✓ Specific Ground Range (SGR) is defined as the ground range (gnm) covered per Kg of fuel used. The higher the SGR the more economical the flight.

To calculate ground nautical miles per Kg:

$$\boxed{\mathbf{SGR} = \frac{\mathbf{Groundspeed(gnm/hr)}}{\mathbf{Fuel\ Flow\ (Kgs/hr)}}}$$

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CRITICAL POINT (CP) / POINT OF EQUAL TIME (PET) EQUITIME POINT

- ✓ The point along the track from which flying time to two selected bases will be same.
- ✓ Required in event of emergency in the aircraft which requires landing to be made as soon as possible.
  - A. In case of engine failure
  - **B.** Passenger falls seriously sick (Heart Attack)

$$\textbf{DISTANCE TO PET}\ (\mathbf{X}) = \frac{\mathbf{DH}}{\mathbf{O+H}}$$

- ☞ Where D is the total Distance between two bases
- ☞ H is reduced Ground Speed Home (Departure)
- ☞ O is reduced Ground speed Out (Destination)

$$\textbf{TIME TO PET} = \frac{\textbf{Distance to PET}}{\textbf{Normal Ground Speed Out}}$$

Note-

- ☞ Distance to Pet is calculated with reduced Ground speed if single engine.
- ☞ For time to PET Normal Ground speed is used.

IMPORTANT FACTS

- ✓ PET will always be on track and upwind or into the wind
- ✓ PET will normally be somewhere near midpoint when
  1. O=H
  2. Nil winds

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3. Beam winds (winds 90 deg. To track)

- ✓ in case of ↑ HW, dist. to CP or CP shift towards destination. In case of ↑ TN, dist. to CP shifts towards departure.
- ✓ When TAS is reduced, PET will move further into Wind
- ✓ PET is independent of FUEL/ENDURANCE
- ✓ Effect of increasing TAS is same as that of reducing wind and PET always moves towards Departure i.e. Base Airfield
- ✓ In beam winds i.e. 90 degree to track TCP increases as is a little of headwind to crab into the wind and DCP is same as in nil winds since there is a little of headwind.
- ✓ Beam Winds are winds to track
- ✓ If the fuel consumption increases or flight fuel reduces, distance to CP will remain the same.
- ✓ If beam wind doubles, dist. To CP will remain the same but TCP increases

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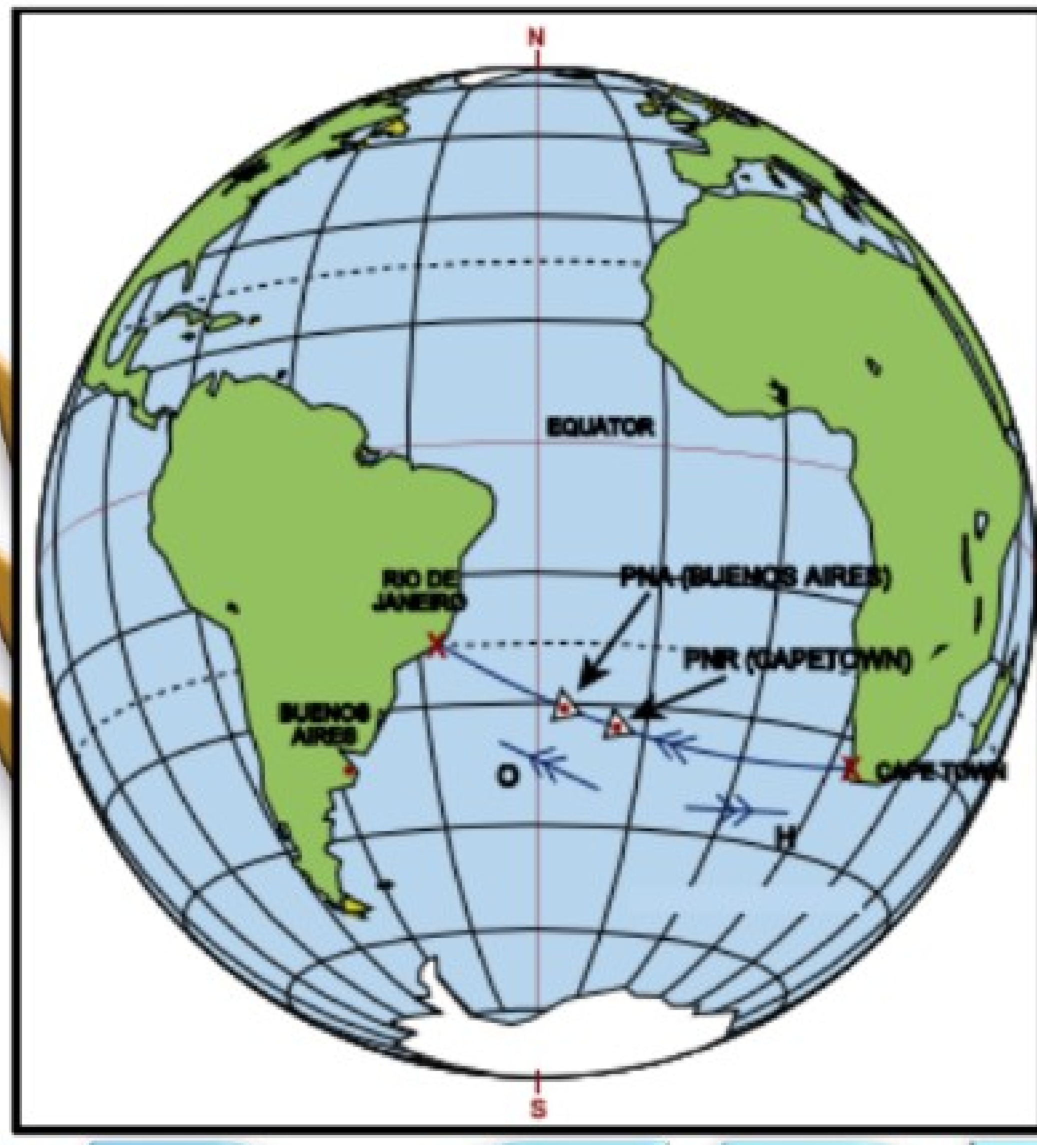
PNR

POINT OF SAFE RETURN (PSR) OR POINT OF NO RETURN (PNR) OR RADIOUS OF ACTION (ROA)



- ✓ PSR is the maximum distance an aircraft can fly out and still be able to return to selected base within SAFE ENDURANCE OF AIRCRAFT
- ✓ Required if both the destination and the destination alternate are not available.

$$\textbf{TIME TO PSR} = \frac{\mathbf{E \times H}}{\mathbf{O + H}}$$

- ☞ Where E is safe endurance
- ☞ H is reduced ground speed home
- ☞ is Normal ground speed out

$$\textbf{DISTANCE TO PNR} = \frac{\textbf{EOH}}{\textbf{O + H}}$$

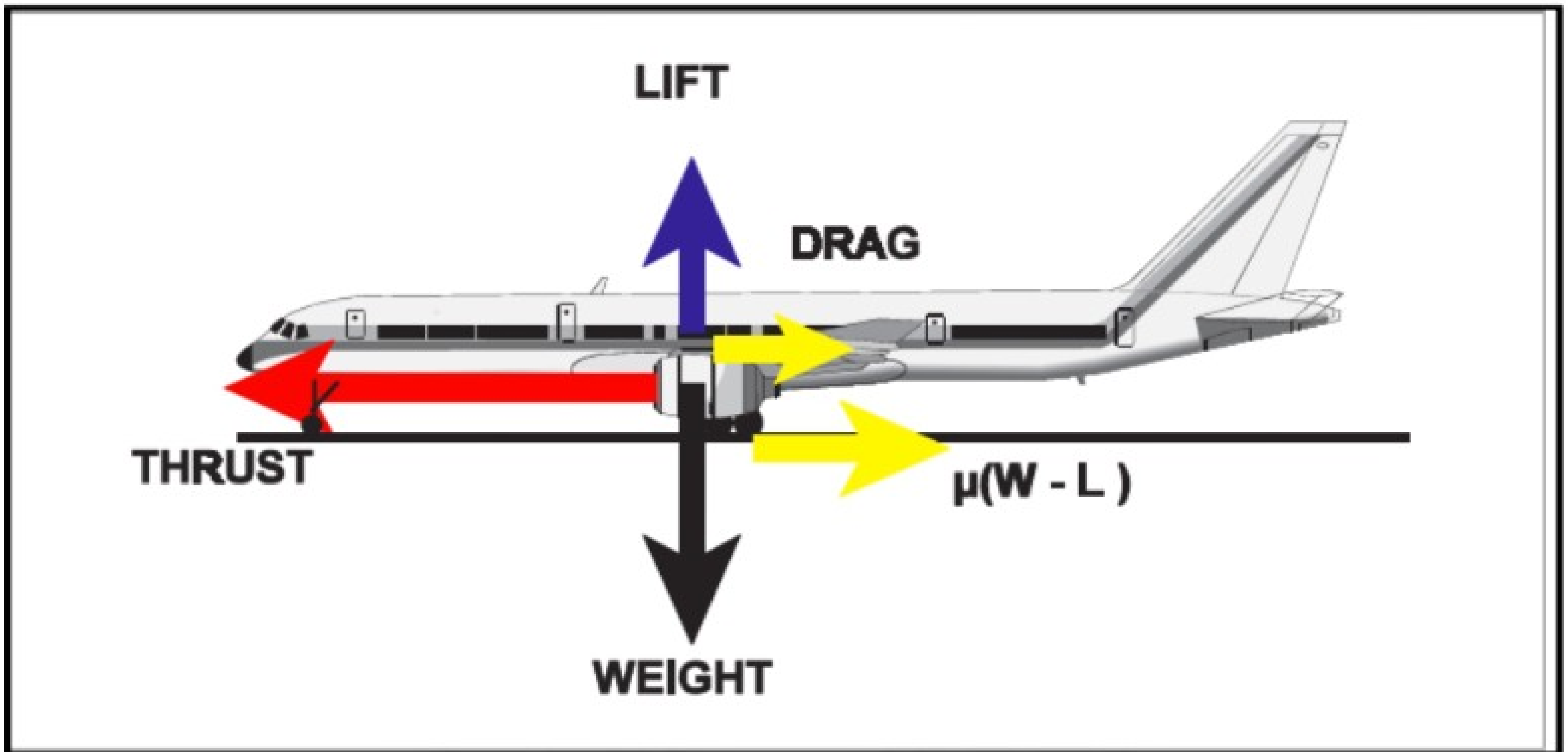
STILL AIR RANGE

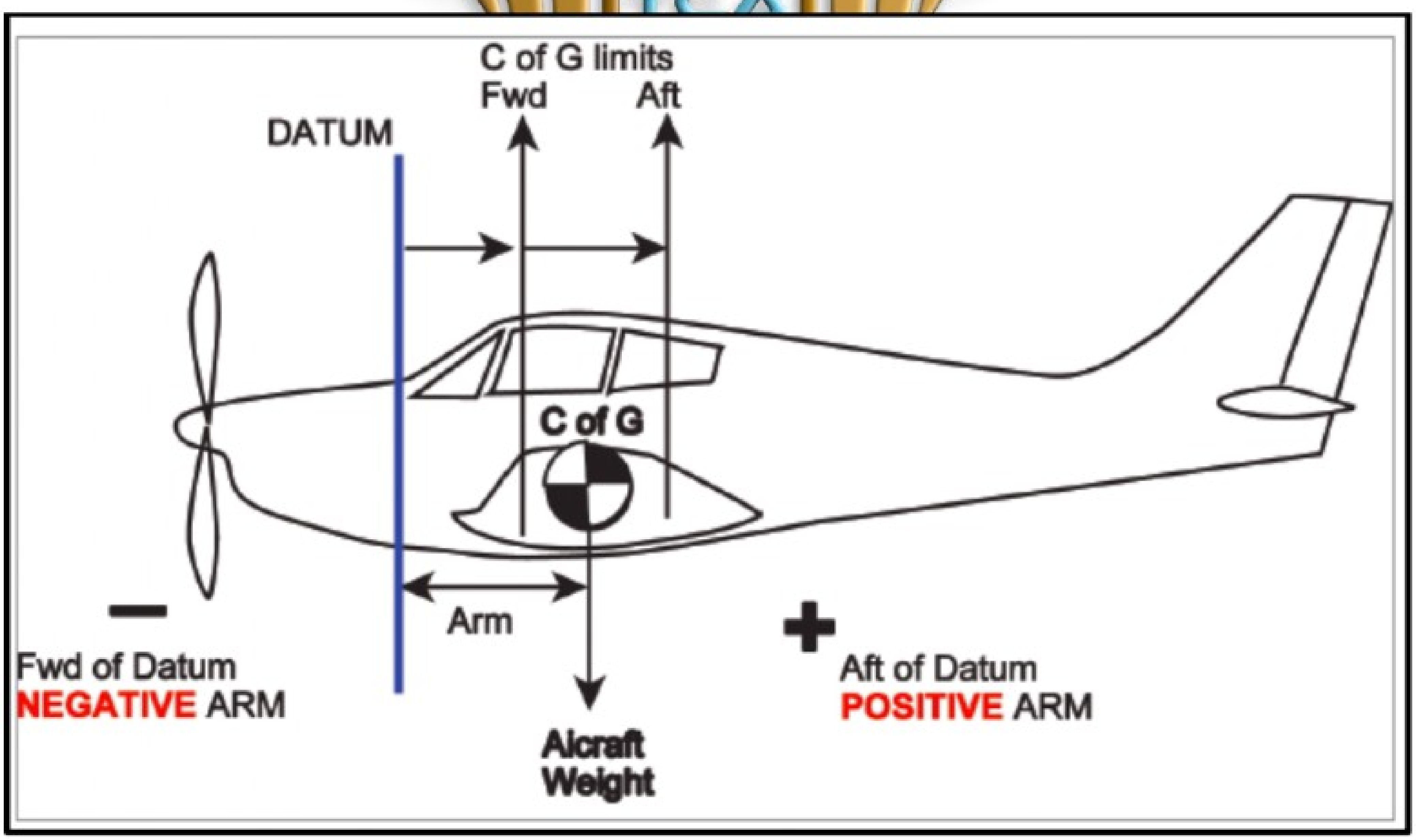
- ✓ Max distance an aircraft can fly in still air
- ✓ Is equal to Total Endurance X TAS

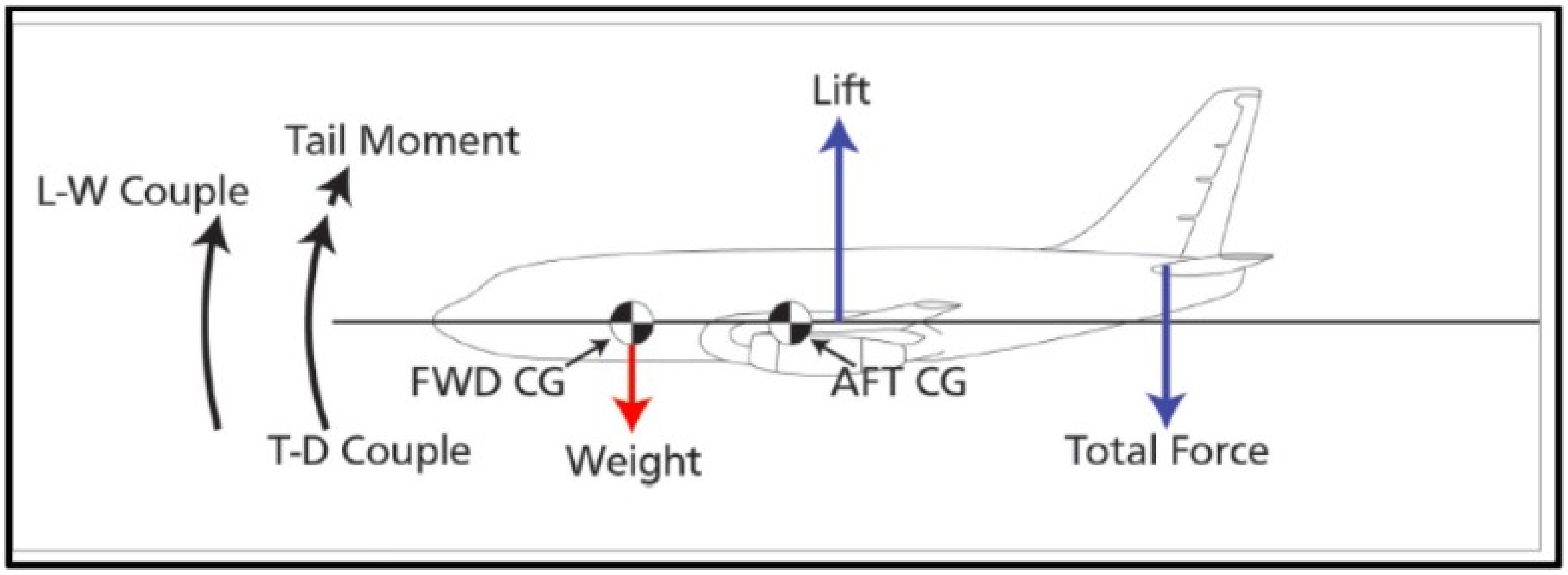
FACTORS AFFECTING PNR (PSR)

1. Greatest distance from departure point to PNR occurs in still air.
2. Any wind component, no matter it is Head or Tail wind will cause PNR to be closer to departure point or distance to PNR will decrease in wind.
3. Abeam wind also reduces distance to PNR.
4. If Headwind changes to tailwind, distance to PNR will remain same.
5. Constant Wind reduces distance to PNR from still air condition.
6. Distance will vary directly with Endurance.
7. Carriage of extra fuel -distance to PNR increases- distance to CP remains same.
8. Any improvement in performance will increase distance to PNR.
9. If fuel flow decreases, distance to PNR will increase.
10. If fuel carried is only Flt fuel + reserve (Safe endurance Flight Time, CP and PNR will coincide.
11. PNR will usually be beyond PET.

PERFORMANCE







Centre of Gravity

- ✓ The point through which the force of gravity is said to act on a mass (in aircraft terms, the point on the aircraft through which the total mass is said to act in a vertically downward manner).
- ✓ The centre of gravity is also the point of balance and as such it affects the stability of the aircraft both on the ground and in the air.

Centre of Gravity Limits

- ✓ The CG is not a fixed point; it has a range of movement between a maximum forward position and a maximum rearward position which is set by the aircraft manufacturer and cannot be exceeded.
- ✓ The CG must be on or within the limit range at all times. The limits are given in the flight manual and are defined relative to the datum.
- ✓ They may also be given as a percentage of the mean chord of the wing. (The wing mean chord was called the Standard Mean Chord but is now known as the Mean Aerodynamic Chord or more simply, the MAC.)

Exceeding CG forward limit

- ✓ Easy to recover from stall.
- ✓ Apparent weight ↑ so stall speed ↑

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- ✓ Aproach speed ↑ and landing distances required ↑
- ✓ Moment ↑ as arm ↑ because **(m = arm X force)**
- ✓ Take-off speeds V1 , VR, VMU will ↑
- ✓ Total drag ↑
- ✓ Fuel consumption ↑ so endurance + range = ↓
- ✓ Longitudinal stability ↑
- ✓ Controllability ↓

Exceeding CG after limit

- ✓ Longitudinal stability is reduced and, if the CG is too far aft, the aircraft will become very unstable (like a bucking bronco).
- ✓ Stick forces in pitch will be light, leading to the possibility of over stressing the aircraft by applying excessive ‘g’.
- ✓ Recovering from a spin may be more difficult as a flat spin is more likely to develop.
- ✓ Range and endurance will probably decrease due to the extra drag caused by the extreme manoeuvres. Glide angle may be more difficult to sustain because of the tendency for the aircraft to pitch up.
- ✓ Controllability ↑

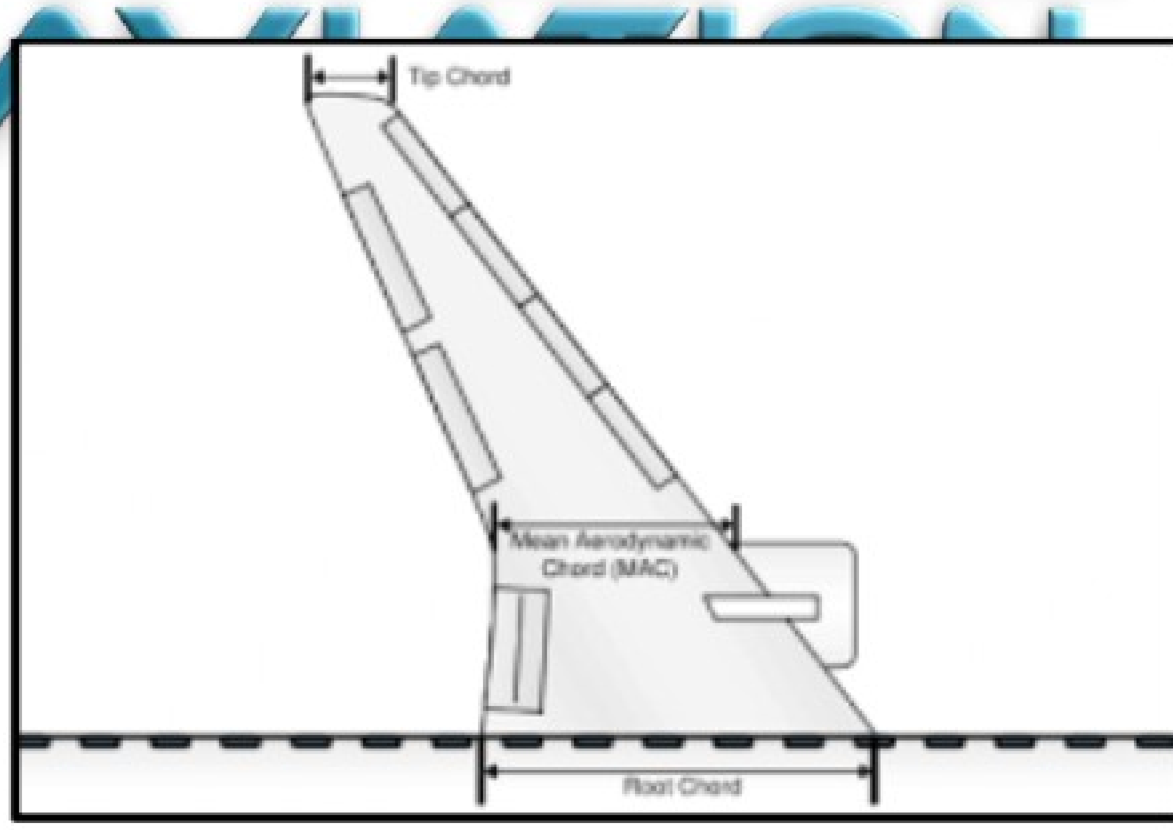
CG position moves during flight due:

1. Fuel usages
2. Flap + landing gear operations
3. Crew + Passenger movement

CG ON FWD LIMIT		CG ON AFT LIMIT	
STABILITY	↑	STABILITY	↓
STICK FORCES	↑	STICK FORCES	↓
MANOEUVRABILITY	↓	MANOEUVRABILITY	↑
DRAG	↑	DRAG	↓
V_S (STALLING SPEED)	↑	V_S	↓
V_R (ROTATION SPEED)	↑	V_R	↓
RANGE	↓	RANGE	↑
FUEL CONSUMPTION	↑	FUEL CONSUMPTION	↓
ABILITY TO ACHIEVE		**ABILITY TO ACHIEVE**	
1. CLIMB GRADIENT	↓	CLIMB GRADIENT	↑
2. GLIDE SLOPE	↓	GLIDE SLOPE	↑

MAC (Mean Aerodynamic Chord)

- ✓ Mean Aerodynamic Chord is the average chord length of a tapered, swept wing.
- ✓ The Length of the chord of a rectangular wing with same area and aerodynamic properties as original wing.
- ✓ In large aircraft, centre of gravity limitations and the actual centre of gravity are often expressed in terms of percent MAC.



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Chord

- ✓ The distance between the leading and trailing edge of the wing, measured parallel to the normal airflow over the wing, is known as the chord.

Datum

- ✓ A point along the longitudinal axis (centre line) of the aeroplane (or its extension) designated by the manufacturer as the zero or reference point from which all balance arms (distances) begin.
- ✓ By taking moments about the datum the CG position of the aircraft can be determined. For the purposes of this phase of study the lateral displacement of the CG from the longitudinal axis is assumed to be zero.

Moment

- ✓ Turning effect of mass around datum.
- ✓ Moment = **Balance arm x Mass**
- ✓ Standard mass – for flight crew is 85kg and for cabin crew is 75kg.
- ✓ Aircraft weighing is carried out in a fully enclosed non air conditioned place.

$$\boxed{\textbf{CG arm} = \frac{\textbf{Total moment}}{\textbf{Total mass}}}$$

Balance Arm

- ✓ The distance from the aircraft's datum to the CG position or centroid of a body or mass.
- ✓ For example, the centroid of a square or rectangle is the exact centre of the square or rectangle and, in such cases; the balance arm is the distance from the datum to the exact centre of the square or rectangle.

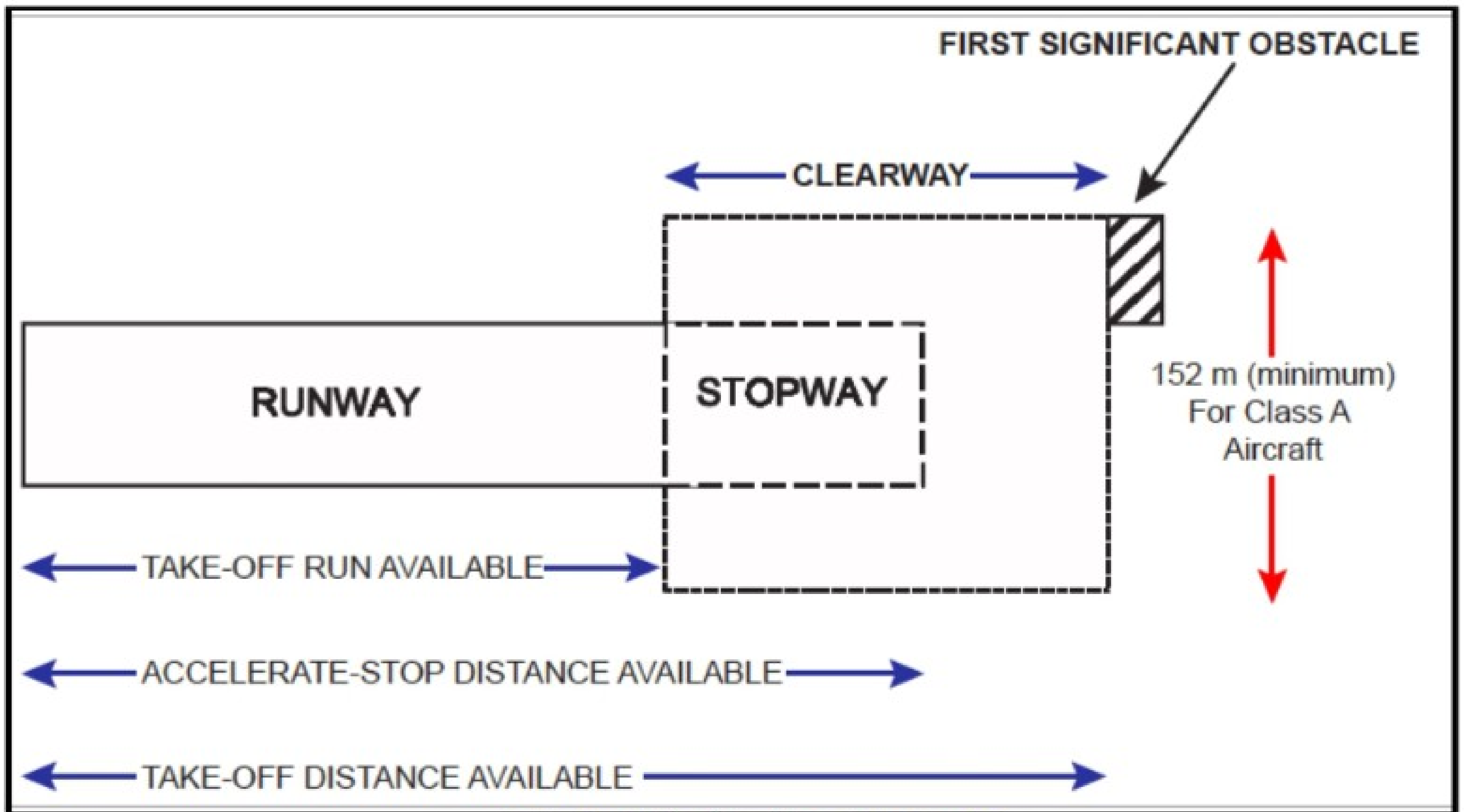
For the purposes of calculations, all balance arms ahead of (in front of) the datum are given a negative (-) prefix and those behind (aft of) the datum are given a positive (+) prefix

- Datum line is on the longitudinal axis but not necessary between nose and tail
- Distance between datum line and cg is called the balance arm or moment arm
- Stalling speed will be highest with higher mass and fwd cg
- a/c is neutral stable when the cg is at neutral point
- Neutral point is rear of the aft limit
- Degree of stability depends upon distance cg is forward of neutral point

Some Effects of Increasing Aeroplane Mass

V_1	(DECISION SPEED)	$\uparrow$
V_R	(ROTATION SPEED)	$\uparrow$
V_{MU}	(MIN UNSTICK SPEED)	$\uparrow$
V_S	(STALLING SPEED)	$\uparrow$
TAKE-OFF AND LANDING RUN		$\uparrow$
RANGE AND ENDURANCE		$\downarrow$
RATE OF DESCENT		$\uparrow$
MAX HORIZONTAL SPEED		$\downarrow$
RATE OF CLIMB		$\downarrow$
MAX ALTITUDE		$\downarrow$
FUEL CONSUMPTION		$\uparrow$
BRAKING ENERGY		$\uparrow$
TYRE WEAR		$\uparrow$
STRUCTURAL FATIGUE		$\uparrow$

TAKEOFF PERFORMANCE



TAKE OFF

- ✓ The take-off path of the flight is the distance from the brake release point to the point at which the aircraft reaches a 'screen' of defined height.
- ✓ The available distances at an aerodrome.

THE TAKE-OFF RUN AVAILABLE (TORA)

- ✓ The length of runway suitable for normal operations.

THE TAKE-OFF DISTANCE AVAILABLE. (TODA)

- ✓ The length of runway plus any clearway. **(TORA +CLEARWAY)**

THE ACCELERATE-STOP DISTANCE AVAILABLE (ASDA)

- ✓ Take-off Run Available+ Stop way **(TORA+STOPWAY)**

CLEARWAY

- ✓ It is a rectangular area with defined dimension at the end of runway free from the obstacles either over the water or over the land and used only for take off.
- ✓ The aircraft must achived the screen height at the end of the clear way.

NOTE: Minimum width of the clearway is 150 mtr and maximum gradient should not exceed 1.25%.

STOPWAY

- ✓ It is a rectangular area at the end of the runway defined dimension used only for take-off.
- ✓ And centered upon the extended centerline of the runway, able to support the aero plane during an abortive take-off, without causing structural damage to the aero plane, and designated by the airport authorities for use in decelerating the aero plane during an abortive take-off.

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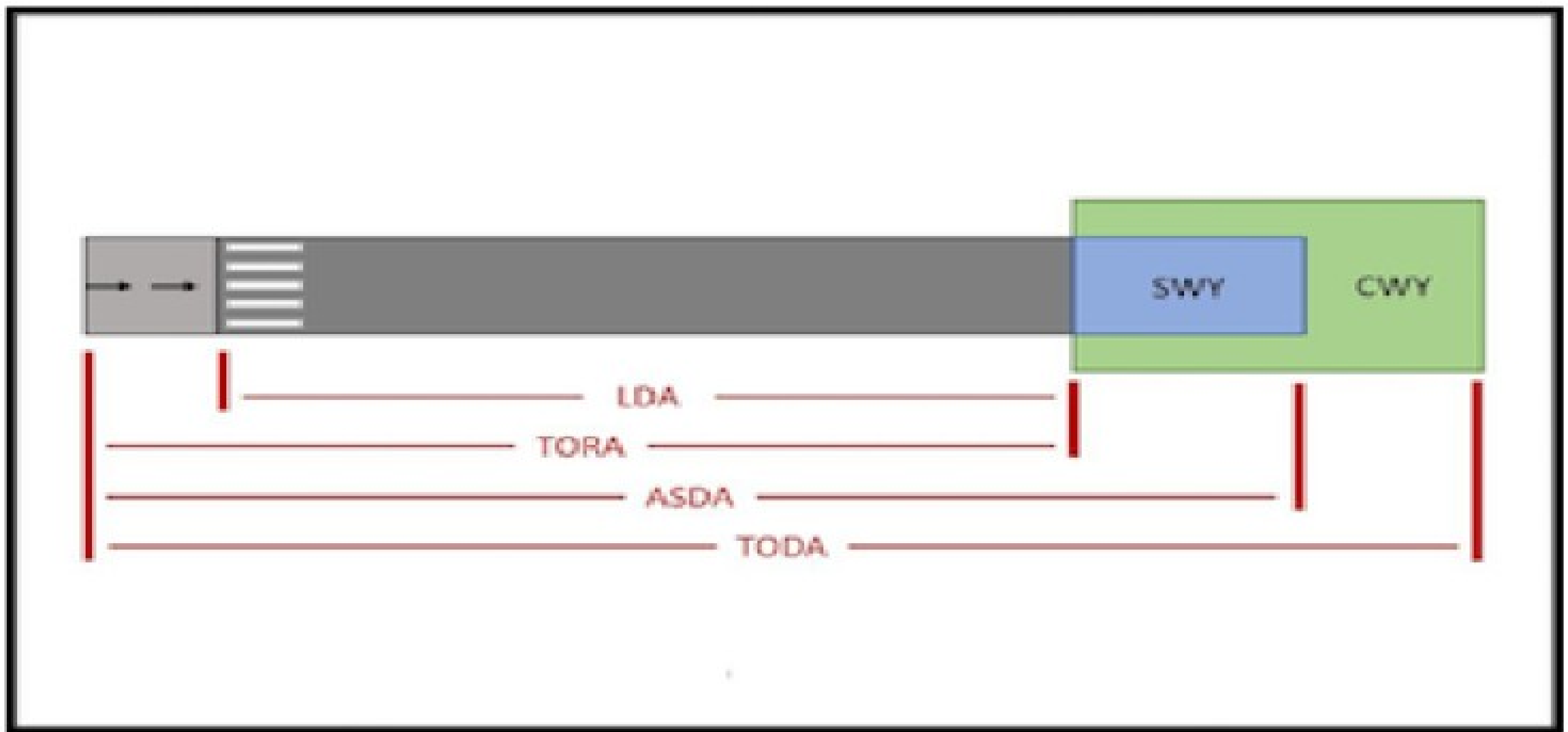


SCREEN HEIGHT

- ✓ The take off part of the flight is the distance from the brake release point (BRP) to the point at which the aircraft reaches a defined height. This defined height is termed the "screen height".
- ✓ The screen height **varies from 35 ft for class A aeroplanes to 50 ft for class B aeroplanes.**

DRAG

- ✓ The total drag (D) of an aeroplane during take-off is a product of both aerodynamic drag (DA) and wheel drag/wheel friction (μ) as shown in the formula below.

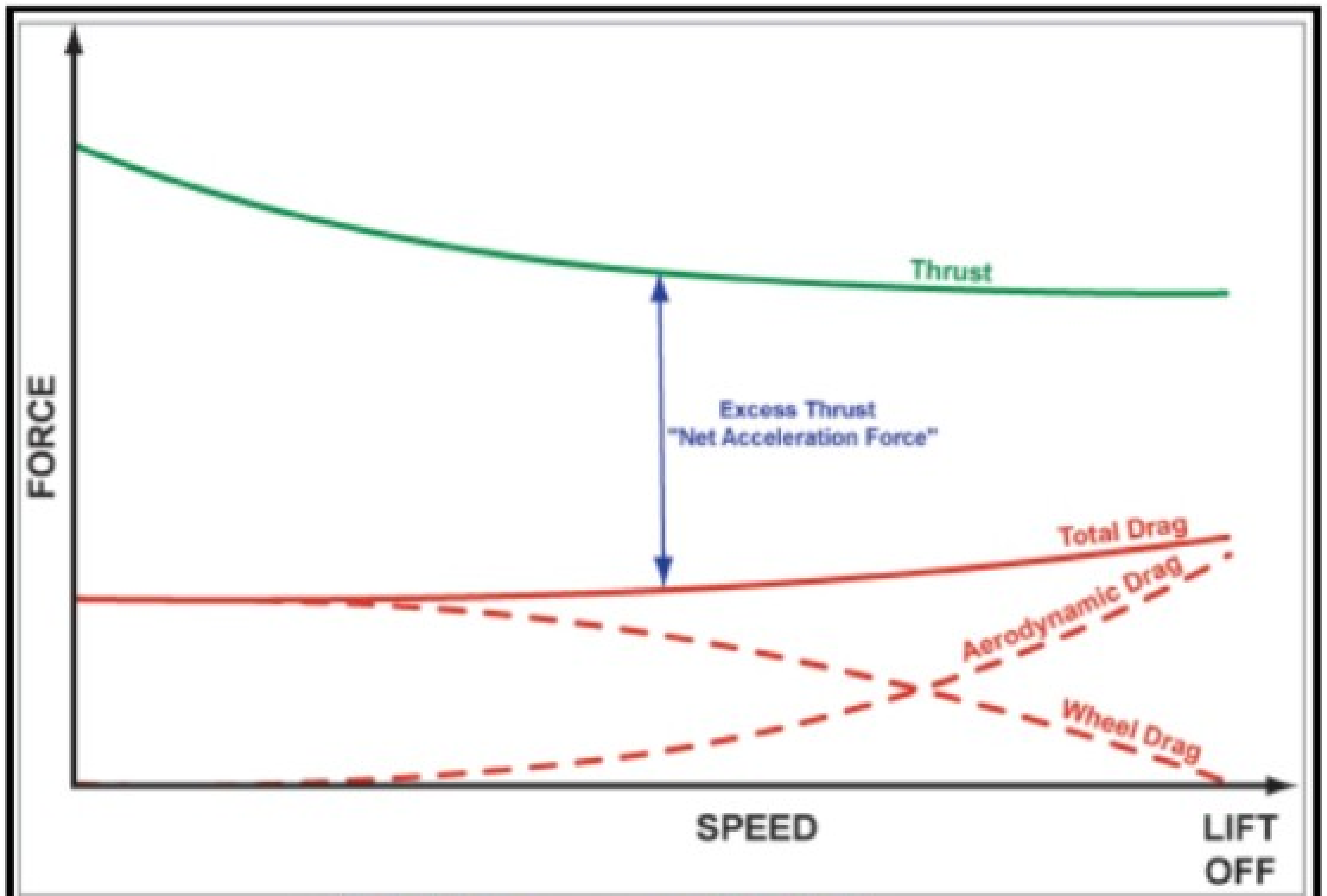
$$\boxed{D = DA + \mu(W - L)}$$

AERODYNAMIC DRAG

- ✓ There are principally two forms of aerodynamic drag,
  1. Parasite drag
  2. Induced drag.

Parasite drag - is increased by the square of the speed, therefore this form of drag will increase during the take-off.

Induced drag - is a function of the angle of attack. This angle of attack is constant until the aeroplane rotates at which point the angle of attack increases dramatically. Therefore induced drag will increase during the take-off.



VIMD / VMD (Only for jet engine) Minimum drag

- ✓ Best glide range
- ✓ Minimum drag speed
- ✓ Maximum endurance speed
- ✓ Best angle of climb
- ✓ Best L/D ratio
- ✓ VX (Best angle of climb speed) lies at VMD

Wheel Drag

- ✓ It is a function of cofficient of friction and load plays on wheels

Wheel drag = μ(W - L) (W =Weight, L= Lift)

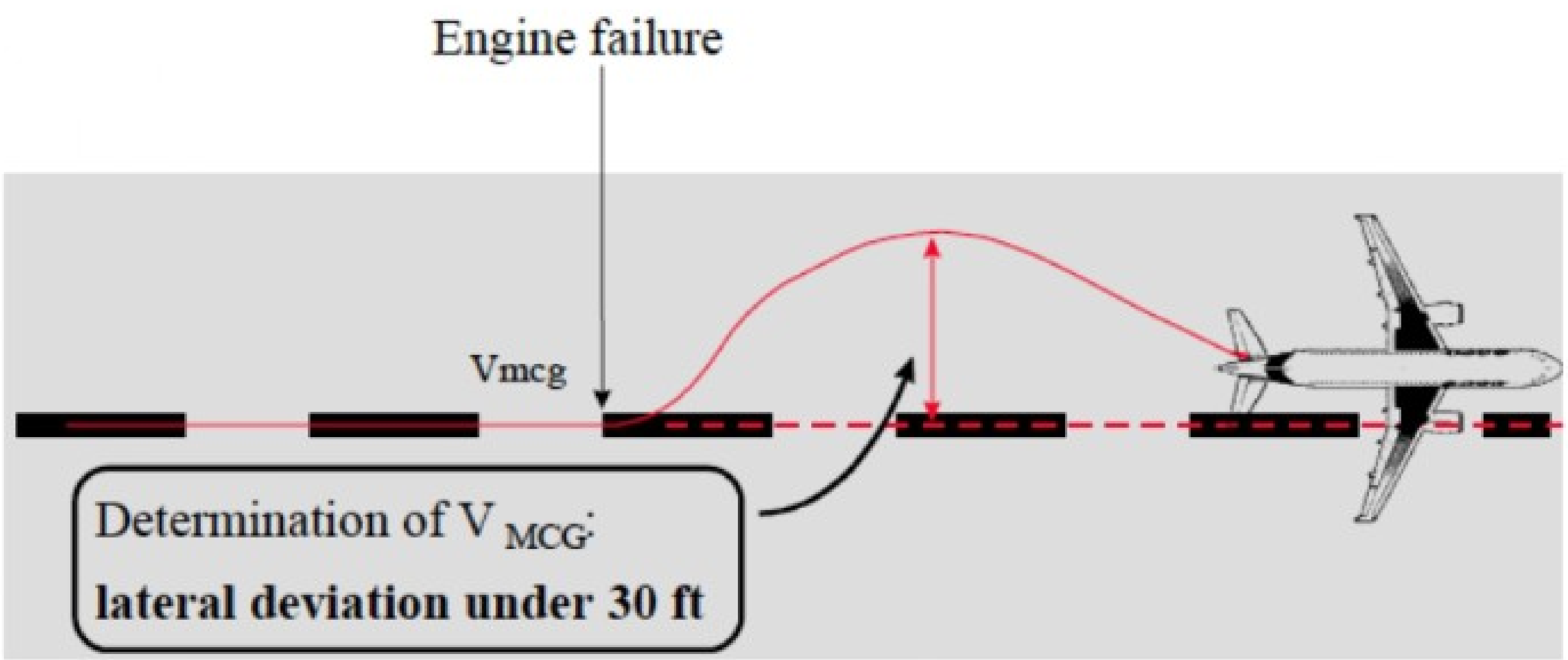
TAKE OFF SPEEDS

VMCG (Minimum Controllable speed Ground)

- ✓ VMCG, the minimum control speed on the ground, is the calibrated airspeed during the take-off run, at which, when the critical engine is suddenly made inoperative.
- ✓ It is possible to maintain control of the aeroplane with the use of the primary aerodynamic controls alone (without the use of nose-wheel steering) to enable the take-off to be safely continued using normal piloting skill.
- ✓ In the determination of VMCG, assuming that the path of the aeroplane accelerating with all engines operating is along the centreline of the runway
- ✓ Its path from the point at which the critical engine is made inoperative to the point at which recovery to a direction parallel to the centreline is completed, may not deviate more than 30 ft laterally from the centreline at any point.

NOTE:

- The thrust produced by the live engine is one of the most important factor for the VMCG.
- Greater the thrust of the live engine VMCG would be higher.
- Maximum rudder pressure is 150lbs.

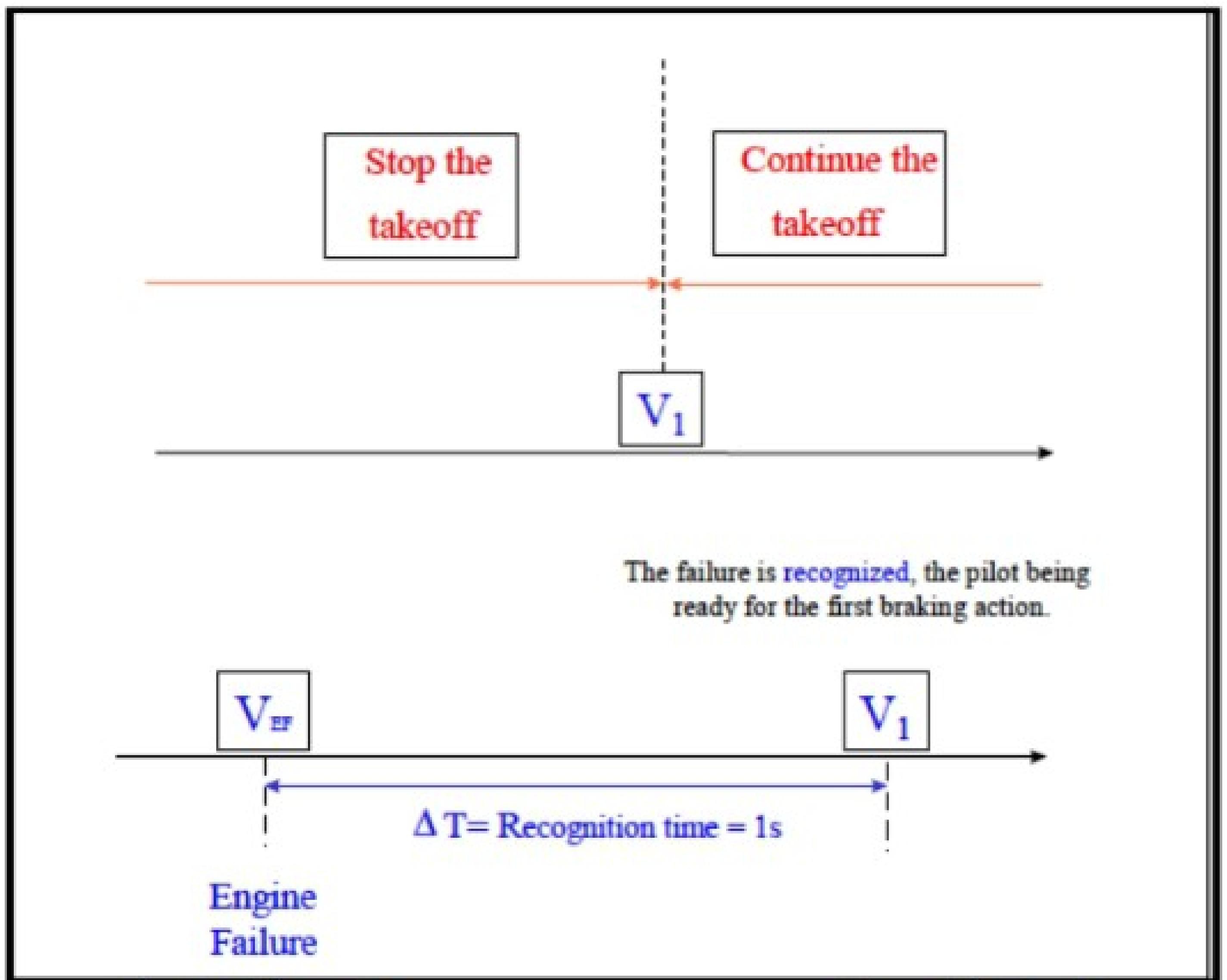


VEF (assumed engine failure speed)

- ✓ VEF is the calibrated airspeed at which the critical engine is assumed to fail.
- ✓ VEF must be selected by the applicant, but may not be less than VMCG."
- ✓ It is more of the manufacturer use not for the operational use.

V1 (Decision Speed)

- ✓ V1 is defined as the maximum speed at which the pilot must take the first action to stop the aeroplane within the remaining accelerated stop distance.
- ✓ V1 is also the minimum speed following engine failure that the pilot is able to continue the take-off within the remaining take-off distance.
- ✓ **Engine failure prior to V1** demands that the pilot must abandon the takeoff because there is insufficient distance remaining to enable the aircraft accelerate to the screen height.
- ✓ **Engine failure after V1** demands that the pilot must continue the take-off because there is insufficient distance remaining to safely bring the aircraft to a stop.
- ✓ **VMCG ≤ VEF ≤ V1**



VR (Rotation Speed)

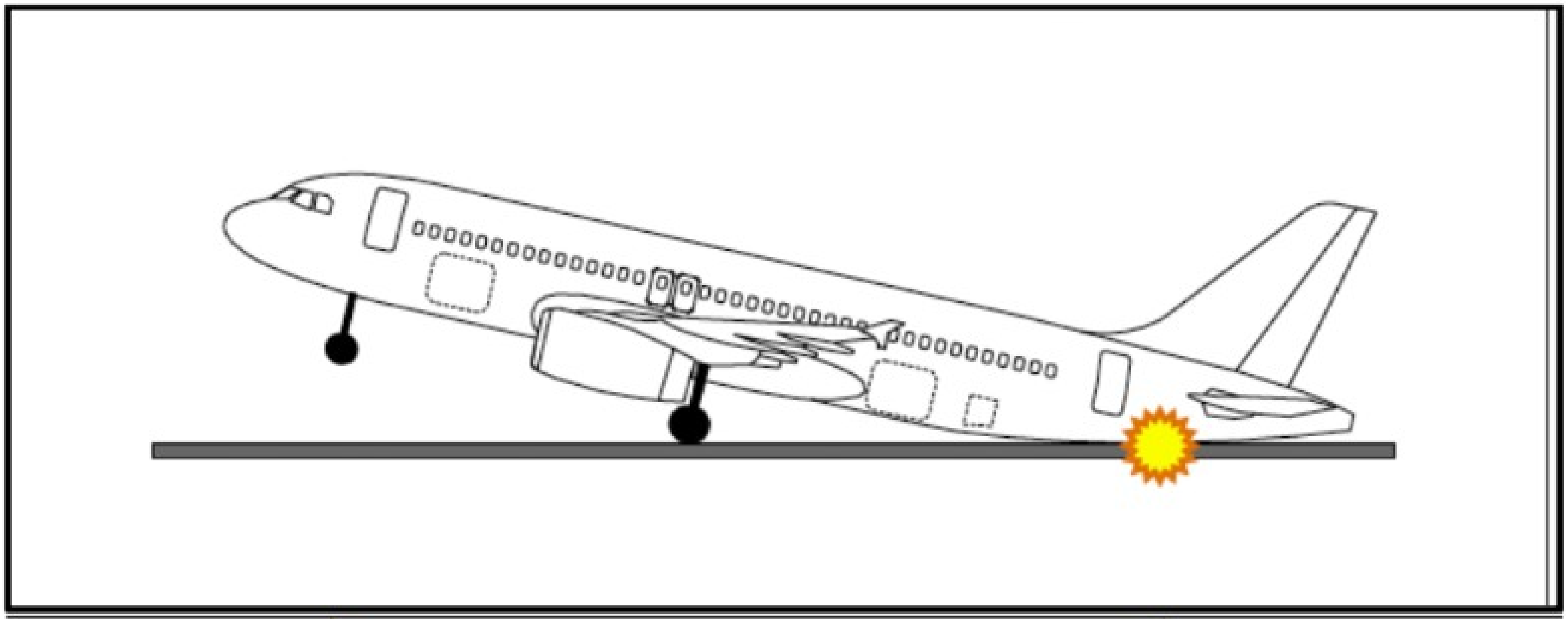
- ✓ VR, in terms of calibrated airspeed, may not be less than:V1, 105% of VMCA
- ✓ The speed that allows reaching V2 before reaching a height of 35 ft above the take-off surface, or
- ✓ A speed that, if the aeroplane is rotated at its maximum practicable rate, will result in a [satisfactory] VLOF"
- ✓ VR ≥ 1.05 VMCA

VMBE (Maximum brake-energy speed)

- ✓ When the takeoff is aborted, brakes must absorb and dissipate the heat corresponding to the aircraft's kinetic energy at the decision point
- ✓ V1 ≤ VMBE

VMU (The minimum unstick speed)

- ✓ VMU is the calibrated airspeed at and above which the aeroplane can safely lift off the ground, and continue the take-off.



VLOF (Lift-off Speed)

- ✓ VLOF is the calibrated airspeed at which the aeroplane first becomes airborne. Therefore, it is the speed at which the lift overcomes the weight.
- ✓ VLOF must not be less than 110% of VMU in the all-engines-operating condition.
- ✓ And not less than 105% of VMU determined at the thrust-to-weight ratio corresponding to the one-engine-inoperative condition.

VTIRE (Maximum Tire Speed)

- ✓ The tire manufacturer specifies the maximum ground speed that can be reached, in order to limit the centrifugal forces and the heat elevation that may damage the tire structure. Thus:
- ✓ VLOF ≤ VTIRE

V2 (Take-Off Safety Speed)

- ✓ V2 is the minimum climb speed that must be reached at a height of 35 feet above the runway surface, in case of an engine failure.
- ✓ **Take-off safety speed A** referenced airspeed obtained afterlift-off at which the required one engine-inoperative climb performance can beachieved.

VA - Design Maneuvering Speed.

VFE - The maximum flap extended speed.

VMD - Velocity of Minimum Drag.

VMP - Velocity of Minimum Power.

VLE - The maximum speed-with landing gear extended.

Landing gear extended speed means the maximum speed at which an aircraft can be safely flown with the landing gear extended.

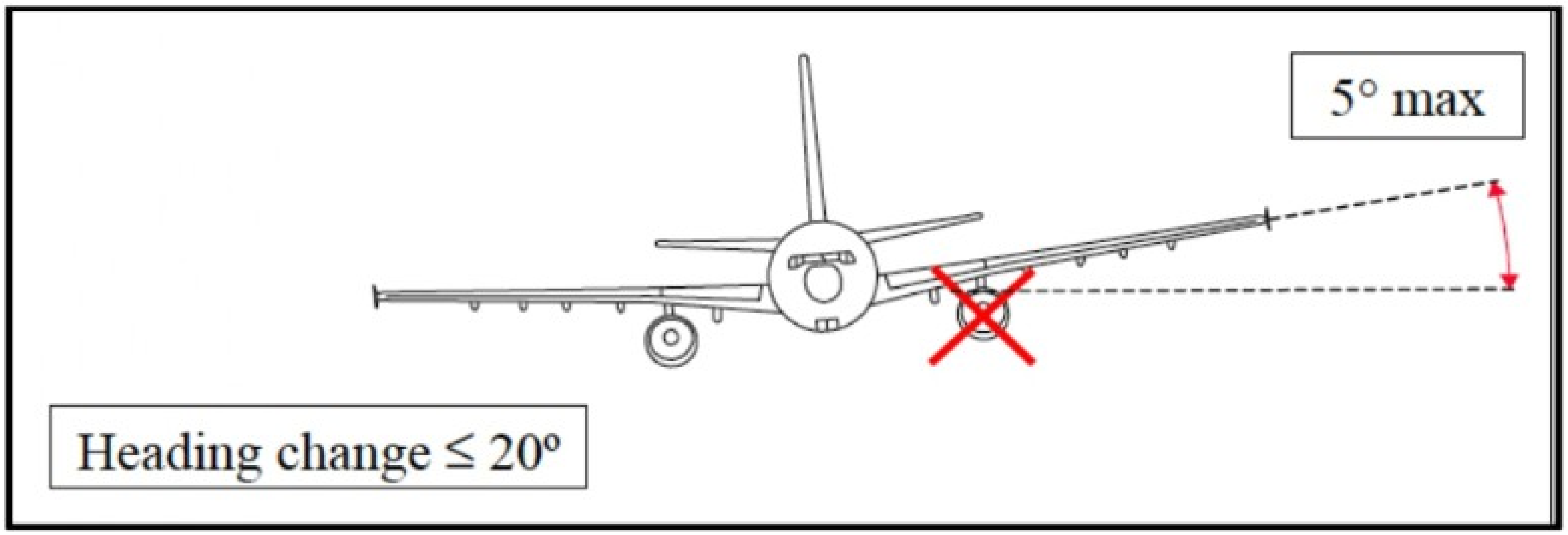
VLO - The maximum at which the landing gear may be lowered.

Landing gear operating speed means the maximum speed at which the landing gear can be safely extended or retracted.

VMC - Minimum control speed with the critical power unit inoperative

VMCA (Minimum controllable speed in the air) (take-off climb)

VMCA - is the calibrated airspeed, at which, when the critical engine is suddenly made inoperative, it is possible to maintain control of the aeroplane with that engine still inoperative, and maintain straight flight with an angle of bank of not more than 5 degrees.



V_{MCL} - Landing minimum control speed (on the approach to land)

V_{MO} - The maximum operating speed

V_{NE} - Never exceed speed

V_{REF} - The reference landing speed (Replaced VAT Speed) **VS** Stalling speed or minimum steady flight speed at which the aeroplane is Controllable **1.30 V_{SO}**

V_{S1G} - Stalling speed at 1g or the one-g stall speed at which the aeroplane can develop a lift force (normal to the flight path) equal to its weight. This is assumed to be the same speed as VSR.

V_{SO} - The stalling speed with the flaps at the landing setting or minimum steady flight speed at which the aeroplane is controllable in the landing configuration

V_{S1} - The stalling speed for the configuration under consideration

V_X - The speed for the-best gradient or angle of climb

V_Y - The speed for the best rate of climb

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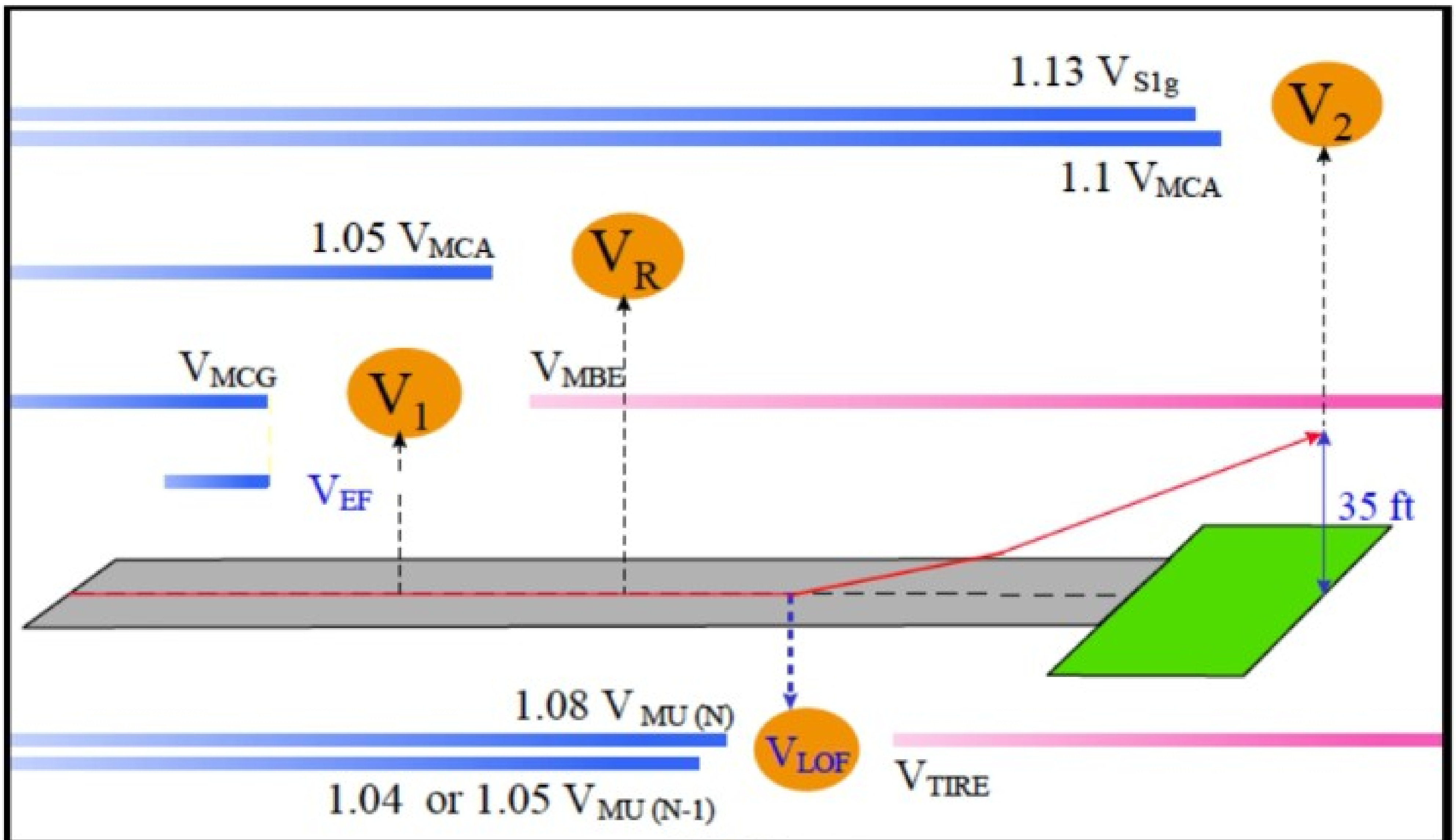
Speed Summary

✓ The following Figure illustrates the relationships and the regulatory margins between the certified speeds (VS1G, VMCG, VMCA, VMU, VMBE, VTIRE), and the takeoff operating speeds (V1, VR, VLOF, V2)



PERFORMANCE DEFINITIONS

ALTERNATE AIRPORT

- ✓ Means an airport at which an aircraft may land if a landing at the intended airport becomes inadvisable.

ASDR/ASDA (Accelerate-Stop Distance Available)

- ✓ ASDR is a destance required from break released to complete stock in case of rejected take off assumed.
- ✓ A/C accelarate with all engine at take of thrust till the point of the engine failure, thereafter the remaining engine is on the take off thrust.
- ✓ Engine failure is recognised at V1 and action initiated to abandoning take off.
- ✓ Then the A/C retart with available means off retardation but without the reverse thurst.

NOTE: In all performance calculations we consider 50% of headwind component and 150% of tail wind component.

BALANCED FIELD

- ✓ Means that **(TODA = ASDA)**
- ✓ Balance field simplify the calculation and enhances the seafty margin.

CALIBRATED AIRSPEED

- ✓ Means indicated airspeed of an aircraft, corrected for position and instrument error. Calibrated airspeed is equal to true airspeed in standard atmosphere at sea level.

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CEILING

- ✓ With respect to aircraft performance, a **ceiling** is the maximum density altitude an aircraft can reach under a set of conditions, as determined by its flight envelope.

Absolute ceiling - Means the pressure altitude at which the rate of climb is zero.

Service ceiling - Means the pressure altitude at which the rate of climb is a defined value.

CLIMB GRADIENT

- ✓ Means the ratio, in the same units, and expressed as a percentage, of: **Change in Height** Horizontal Distance Travelled.

CRITICAL ENGINE

- ✓ Means the engine whose failure would most adversely affect the performance or handling qualities of an aircraft.

DENSITY ALTITUDE

- ✓ Means the height in the International Standard Atmosphere at which the prevailing density occurs.

GROSS PERFORMANCE

- ✓ Means the average performance which a fleet of aeroplanes should achieve if satisfactorily maintained and flown in accordance with the techniques described in the manual.

NET PERFORMANCE

- ✓ Means the gross performance diminished by a margin laid down by the appropriate Authority.

NOTE: Net Performance is always inferior to gross performance. While calculating the net performance the performance degradation factors are taken into account such as:

1. Various in operating techniques
2. Degradation of performance due to aging fleet
3. Deviation of atmospheric condition then assume

PRESSURE ALTITUDE

- ✓ Means the height in the International Standard Atmosphere at which the prevailing pressure occurs; obtained by setting the sub scale of a pressure altimeter to 1013.2 mb(29.92 in 760 mm mercury).

RUNWAY SLOPE

- ✓ Means the difference in height between two points on the aerodrome surface divided by the distance between those points, in the same units, and expressed as a Percentage.

TAKE-OFF DISTANCE AVAILABLE

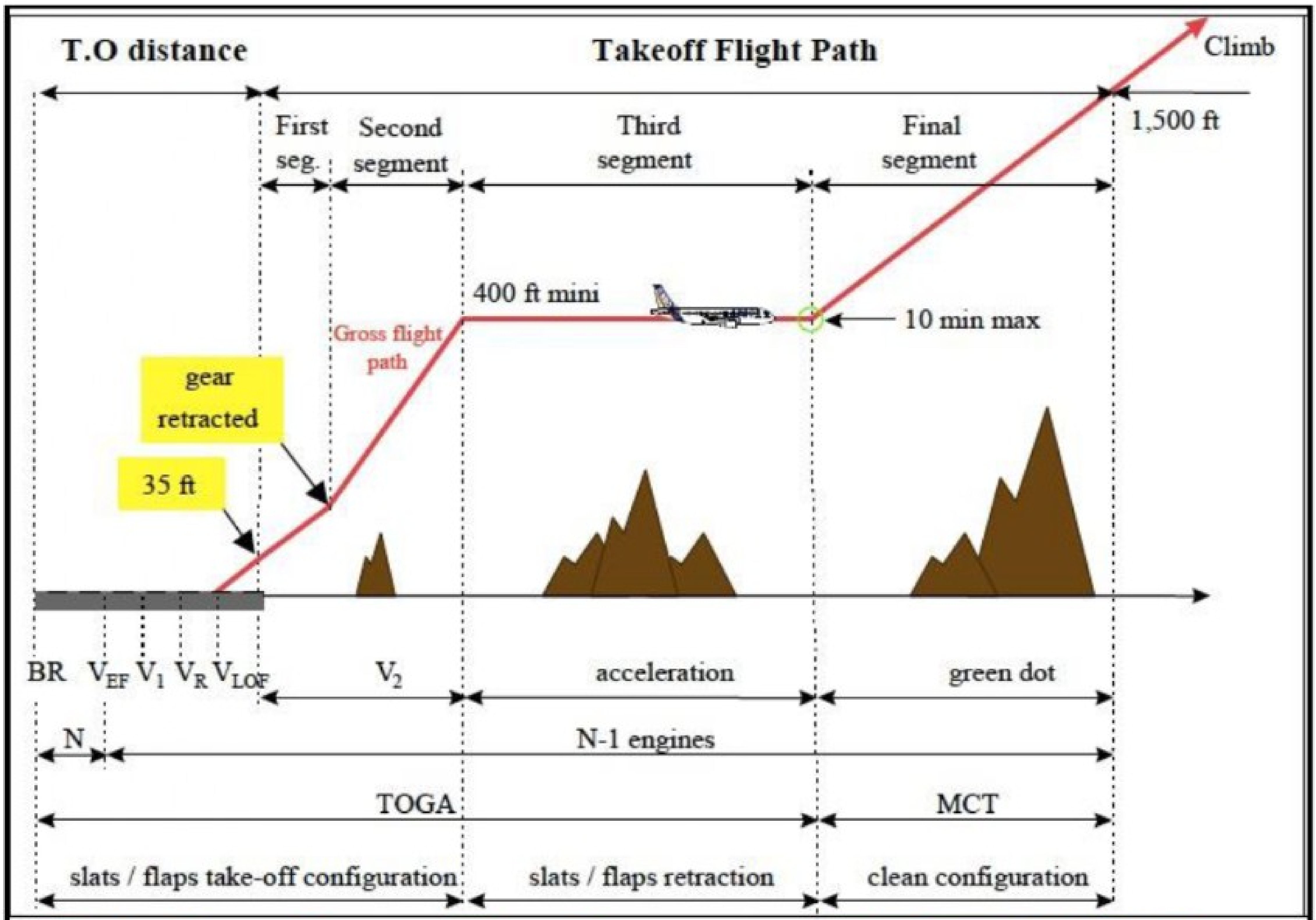
- ✓ Means Take-off Run Available +Clearway

TAKE-OFF SAFETY SPEED

- ✓ Means a referenced airspeed obtained after lift-off at which the required one-engine-inoperative climb performance can be achieved.
- ✓ True airspeed means the airspeed of an aircraft relative to undisturbed air.

		Propeller Driven	
	Multi-engine Jet	**Multi-Engine Turbo-Prop**	**Piston**
Weight: Greater than 5700 kg. Pass: More than 9	A	A	C
Weight: 5700 kg or less Pass: 9 or less	A	B	B

TAKE OFF PERFORMANCE



SEGMENT 1 - Starts at **35 ft** and ends at **gear up completed**

1.2 climb gradient

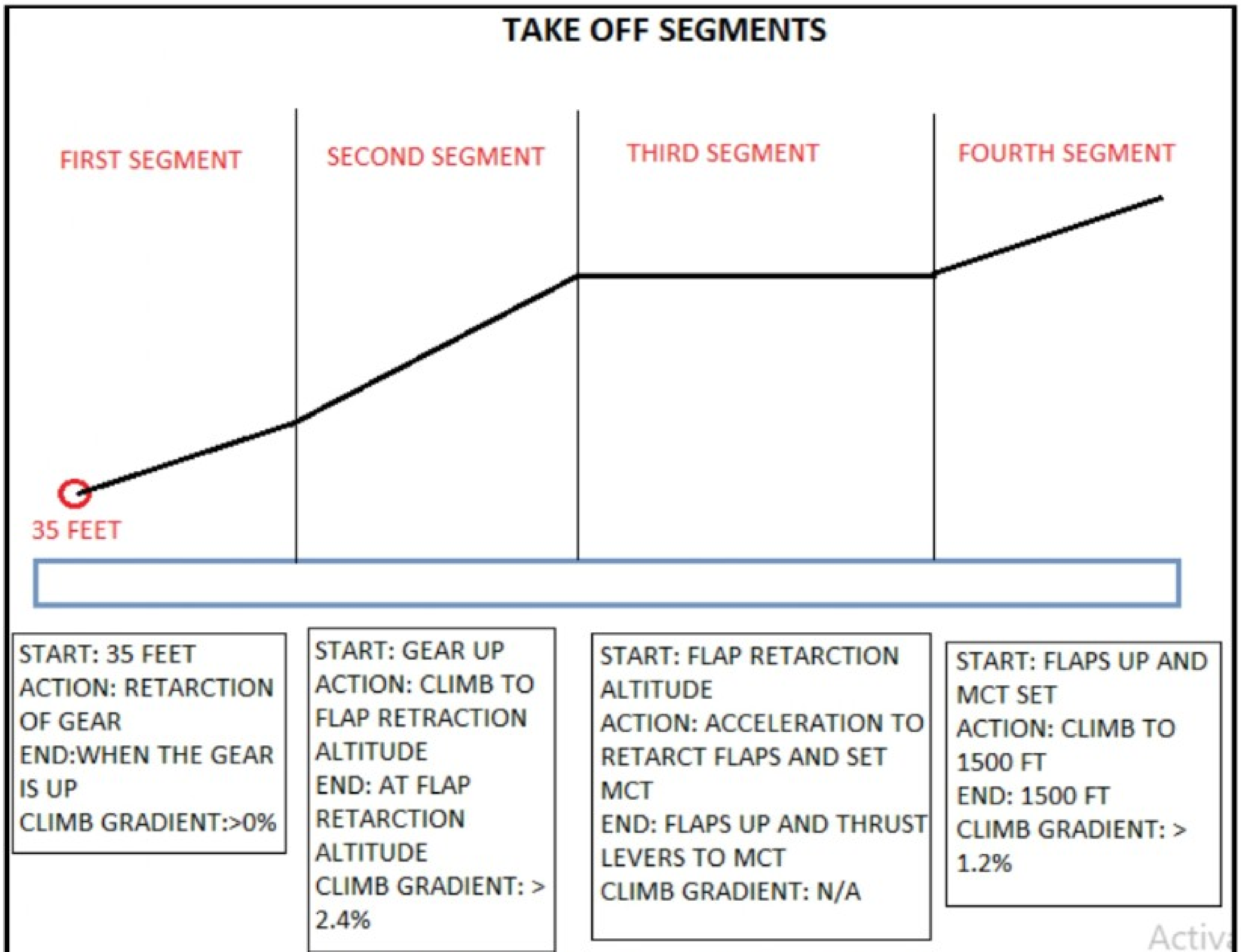
SEGMENT 2 - Starts at **gear up completed** and ends at **400 ft agl** or **acceleration for flap up started**.

2.4

SEGMENT 3 - Starts at **400-ft agl** or **acceleration started** ends at **flap retraction started.**

0

SEGMENT 4 - Starts at **flaps up started** ends at **1500 ft agl.** 1.2



RTR(A) & DGCA GROUND CLASSES

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Cheers and Happy Landings!

Capt P. KUMAR

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